

Phishing Detection Browser Extension

Problem Statement

Phishing websites trick users into entering passwords and personal data on fake pages that look real. Most existing tools either block only already known fake sites using **blacklists**, or give a plain "**safe/unsafe**" verdict with no reasoning behind it. Blacklists always lag behind new attacks because they can only block sites that have already been reported, leaving users unprotected against **zero-day phishing** — newly created fake sites that have not yet been identified. On the other hand, machine-learning-based detectors tend to be more accurate but act as a **black box**, simply saying a site is fake without explaining what made them think so. Since users are far more likely to trust and act on a warning when they understand the reasoning behind it, there is a clear gap between tools that are accurate but unexplained, and tools that are explainable but too slow or complex to run inside a browser.

This project addresses that gap by building a lightweight browser extension that analyzes a website's link in real time using machine learning, and clearly explains **why** a site is considered suspicious — helping users make informed decisions instead of blindly trusting or ignoring a warning.

Summary

The proposed system is a **Chrome extension** backed by a **Random Forest classifier** trained to distinguish phishing links from safe ones, along with an **explainability layer** that turns the model's reasoning into plain language. The overall pipeline is as follows:

- **Data Collection:** Real phishing links are gathered from **PhishTank** and real safe links from **Tranco**, forming the training dataset.
- **Feature Extraction:** Each link is converted into measurable properties such as URL length, number of hyphens/dots, presence of an IP address, use of HTTPS, and subdomain count.
- **Model Training:** A **Random Forest classifier** is trained on these features to learn the patterns that separate phishing links from safe ones.
- **Explainability Layer:** For every flagged link, the specific features that contributed most to the decision are extracted and converted into a plain-language reason.
- **Backend API:** A lightweight server receives a URL from the browser, runs it through the trained model, and returns a risk score along with the reason.
- **Browser Extension:** The user-facing component checks every visited link against the backend and shows a warning popup with the explanation if the site looks suspicious.

The expected outcome is a working Chrome extension that a real user could install, which flags suspicious sites in real time and explains its reasoning — demonstrating the complete pipeline from raw data collection to a deployed, **explainable** security tool that is both **fast** and **accurate** enough to run instantly while browsing.